Zero & Negative Exponents — Practice

Section A is interactive. Sections B and C are written practice — check them against the separate answer key.

Objective (1 mark each)

Q1 · MCQ **[1]**

3^{-2} can be written as:

(a) 9 (b) −9 (c) $1/9$ (d) $-1/9$

Q2 · MCQ **[1]**

The value of $1/4^{-2}$ is:

(a) 16 (b) 8 (c) $1/16$ (d) $1/8$

Q3 · FILL IN THE BLANK **[1]**

For any non-zero integer y, the value of y^0 is:

____

Q4 · MCQ **[1]**

The value of $3^5 \div 3^{-6}$ is:

(a) 3^5 (b) 3^{-6} (c) 3^{11} (d) 3^{-11}

Q5 · TRUE / FALSE **[1]**

$(2/5)^{-1} = 5/2$.

True False

Q6 · ASSERTION–REASON [1]

Assertion (A): The value of 2^{-3} is $1/8$.
Reason (R): For any non-zero integer a, $a^{-m} = 1/a^m$, where m is a positive integer.

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Short answer (2 marks each)

Q7 [2]

Express 16^{-2} as a power with base 2.

Q8 [2]

Evaluate: (i) $(2/5)^{-2}$ (ii) $(-2/3)^4$.

Q9 [2]

Express $27/64$ and $-27/64$ as powers of a rational number.

Long answer (3 marks each)

Q10 [3]

Simplify and write each answer with a positive exponent: (i) $2^{-4} \times 2^7$ (ii) $3^2 \times 3^{-5} \times 3^6$ (iii) $p^3 \times p^{-10}$.

Q11 **[3]**

Express $(-3/4)^{-3}$ with a positive exponent and find its value.

Q12 **[3]**

Simplify $(3^{-2} \times 3^{5}) \div 3^{4}$ and write the answer with a positive exponent.

Total: 21 marks · Check your work against the answer key.